

Rescue carries

A rescue carry is an emergency method of moving a casualty over a short distance to safety, shelter or to transportation. Select the type of carry based on the circumstances.

- The size and weight of the casualty relative to the rescuer
- The number of rescuers available to assist
- The type of injury
- The distance to move the casualty

Whenever possible, ask one or more bystanders to help you. When help is available:

- Remain with the casualty
- Give instructions to the bystanders about what to do and what safety precautions to take
- Fully coordinate the rescue activities

Drag carry

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This carry is used by the single rescuer to drag a casualty who is either lying on their back or in a sitting position. The drag carry provides maximum protection to the head and neck, and therefore should be used when you are moving a casualty with this type of injury.

If time permits, tie the casualty's wrists together across their chest before dragging.



To perform a drag carry:

1. Stand at the casualty's head facing their feet.
2. Crouch down and ease your hands under the casualty's shoulders. Grasp the clothing on each side. Support the casualty's head between your forearms to stop movement.
3. Drag the casualty backward only as far as necessary for their safety.

As an alternate method, the first aider and drag the casualty.



Because of the risk of aggravating any injuries, only use drag carries in the most extreme cases when there is an immediate threat to life.

Human crutch

If a leg or foot is injured, help the casualty to walk on their good leg while you give support to the injured side.



1. Take the weight of the casualty's injured side on your shoulders by placing the casualty's arm (on the injured side) around your neck and grasping the wrist firmly.
2. Reach around the casualty's back with your free hand, and grasp the clothing at the waist.
3. Tell the casualty to step off with you, each using the inside foot. This lets you, the rescuer; take the casualty's weight on the injured side.

Chair carry

The chair carry enables two rescuers to carry a conscious or unconscious casualty through narrow passages and up and down stairs. Do not use this carry for casualties with suspected neck or back injuries. Specially designed rescue chairs are available and should be used for this type of carry.

If the casualty is unconscious or helpless:

1. Place an unconscious casualty on a chair by sliding the back of the chair under their legs and buttocks, and along the lower back.
2. Strap their upper body and arms to the back of the chair.



3. Two rescuers carry the chair, one at the front and one at the back. The rescuer at the back crouches and grasps the back of the chair, while the rescuer at the front crouches between the casualty's knees and grasps the front chair legs near the floor.
4. The rescuers walk out-of-step.

Going down stairs

- The casualty faces forward
- The front rescuer faces the casualty
- A third person should act as a guide and support the front rescuer in case they lose their footing

Extremity carry

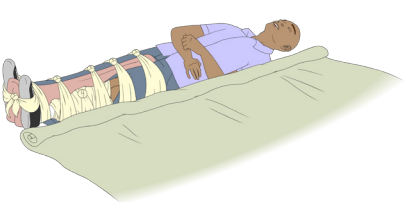
Use the extremity carry when you don't have a chair and do not suspect fractures of the trunk, head, or spine.



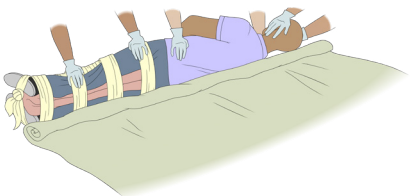
1. One rescuer passes their hands under the casualty's armpits, and grasps the casualty's wrists, crossing them over their chest.
2. The second rescuer crouches with their back between the casualty's knees and grasps each leg just above the knee.
3. The rescuers step off on opposite feet—walking out-of-step is smoother for the casualty.

Blanket lift with four bearers

1. Roll the blanket or rug lengthwise for half its width. Position bearers at the head and feet to keep the head, neck and body in line. Place the rolled edge along the casualty's injured side.

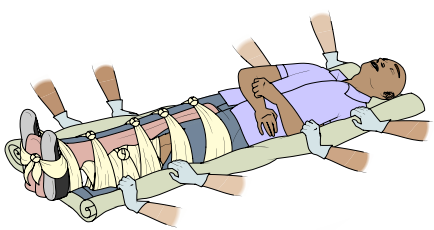


2. Kneel at the casualty's shoulder and position another bearer at the waist to help log-roll the casualty onto the uninjured side. Turn the casualty as a unit so the casualty's body is not twisted.
3. Roll the casualty back over the blanket roll to lay face up on the blanket. Unroll the blanket and then roll the edges of the blanket to each side of the casualty. Get ready to lift the casualty—have the bearers grip the rolls at the head and shoulders, and at the hip and legs.



4. Keep the blanket tight as the casualty is lifted and placed on the stretcher.

Before using a blanket, test it to ensure that it will carry the casualty's weight. Do not use this lift if neck or back injuries are suspected.



Stretchers

There may be times when medical help cannot be contacted, or for other reasons, cannot come to the scene. When this happens, transport the casualty to medical help. If the casualty can't walk, or if the injury or illness allows only the gentlest movement, a stretcher should be used.

Principles for stretcher use

Complete all essential first aid and immobilization before moving the casualty onto a stretcher.

- Bring the blanketed and padded stretcher to the casualty, rather than moving the casualty to the stretcher.
- As the first aider in charge, take the position that permits you to watch and control the most sensitive area of the body, usually at the head and shoulders, or the injured part.

- Tell the bearers what each is expected to do. If the move is difficult, and time permits, it's a good idea to practice with a simulated casualty. This reduces risks and reassures the conscious casualty.
- Test an improvised stretcher with someone equal to or heavier than the casualty to ensure that it will hold.
- Check the clearance of an improvised stretcher to ensure that it will pass through hallways, doors and stairways without harm to the casualty. Use clear commands to ensure smooth, coordinated movements.

Commercial stretchers

The most common of the commercial stretchers is the rigid-pole, canvas stretcher. It has hinged bracing bars at right angles between the rigid poles at either end that must be locked in the extended position before the stretcher is used.



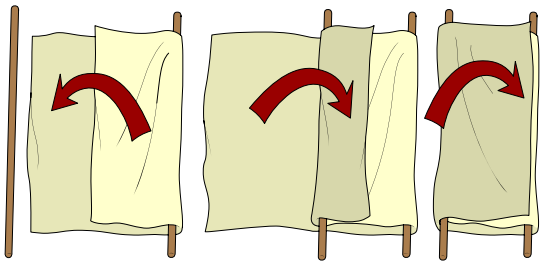
Improvised stretchers

If a commercially prepared stretcher is not available, you can improvise one by using two rigid poles and a blanket, clothing or grain sacks. Do not use non-rigid stretchers for casualties with suspected head or spinal injuries.

Improvised blanket stretcher

1. Place the blanket flat on the ground and place a pole one-third of the way from one end. Fold the one-third length of blanket over the pole.
2. Place the second pole parallel to the first so that it is on the doubled part of the blanket, about 15 cm (6 inches) from the doubled edge.

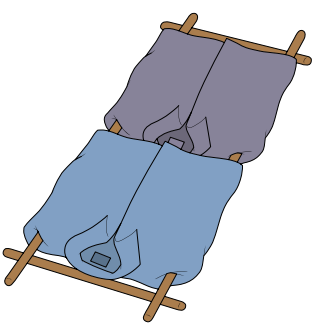
3. Fold the remaining blanket over the two poles. The casualty's weight on the blanket holds the folds in place.



Improvised jacket stretcher

A non-rigid stretcher can also be improvised from two jackets and two or four poles.

1. Button and zipper the jackets closed and pull the sleeves inside out so that the sleeves are inside. Lay the jackets on the ground so that the top edge of one jacket meets the bottom edge of the other.
2. Pass the poles through the sleeves of the two jackets on either side to complete the stretcher.



3. If the casualty is tall, prepare another jacket as before and add it to the stretcher with the head of the jacket towards the middle.

Four-bearer method—no blanket

1. All bearers kneel on their left knees, three on one side of the casualty and one on the other, as shown below. Bearer 4 helps in lifting and lowering the casualty, and also places the stretcher under the casualty.
2. Bearer 4 joins hands with bearers 1 and 2. When assured that each bearer has a firm hold on the casualty, bearer 1 directs the others to "Get ready to lift" and then gives the command "Lift." Lift the casualty smoothly to the height of the raised knees.



3. On bearer 1's command "Rest," the casualty is gently laid on the raised knees of bearers 1, 2 and 3.
4. Bearer 1 tells bearer 4 to position the stretcher. Bearer 4 then resumes their position supporting the casualty by linking their hands with one from each of bearers 1 and 2. Position the stretcher.
5. When everyone is in position, bearer 1 instructs the team to, "Get ready to lower" and then, "lower." The team lowers the casualty gently onto the stretcher. Secure the casualty to the stretcher.



Three-bearer method, no blanket

The three-bearer method is essentially the same as the four-bearer method, except the first aider and one bearer share the weight on one side of the casualty. The third bearer links hands with the first aider from the opposite side to take up the weight of the trunk. The casualty is lifted and rested on the bearers' knees while the stretcher is positioned and bearer 3 links hands again with the first aider to help lower the casualty to the stretcher.

Carrying a stretcher

A stretcher should be carried by four bearers. As the first aider in charge, decide on the carrying method and give clear instructions to the bearers. After the casualty has been strapped to the stretcher, position yourself so you can watch the casualty and at the same time give direction to the other bearers.

Assign the remaining bearers (depending whether you are two or four) to respective corners or ends of the stretcher. Bearers crouch by the carrying handles of the stretcher, facing in the direction of travel.

- When the bearers have a firm footing and a good grip on the stretcher, give the command, "Get ready to lift," and then, "Lift."
- Ask the bearers if they are ready. When they are, give the command, "Go forward."
- When it is necessary to stop, give the commands "Stop," Get ready to lower," and then, "Lower."

To ensure the smoothest carry for the casualty:

- Four bearers carrying a stretcher step off together on the foot nearest the stretcher and keep in step
- Two bearers step off on opposite feet and walk out-of-step

Head-first carry

Although stretcher casualties are usually carried feet first, certain conditions call for a head-first carry:

- Leg injuries during a long downhill carry or when descending stairs, a head-first carry decreases pressure on the lower limbs and minimizes discomfort
- Uphill carries and going up stairs if there are no injuries to the legs—a head-first carry decreases blood flow to the casualty's head and is more comfortable
- Loading an ambulance or transferring the casualty to a bed—it is safer to do this head first, and easier to watch the casualty

Obstacles

When crossing uneven ground, a stretcher should be carried by four bearers and kept as level as possible. Bearers must adjust the height of the stretcher to compensate for dips and rises in the terrain.

Crossing a wall

Avoid crossing a wall, even if it means a longer carry. Where a wall must be crossed, follow these steps:

1. Lift the stretcher onto the wall so that the front handles are just over it. The rear bearers hold the stretcher level while the front bearers cross the wall. All lift together and the stretcher is moved forward until the rear handles rest on the wall.
2. The front bearers hold the stretcher level until the rear bearers have crossed the wall and resumed their positions at the rear of the stretcher.
3. The stretcher is then lowered to continue the journey.

Extrication

Extrication is the process of freeing casualties who are trapped or entangled in a vehicle or collapsed structure and cannot free themselves. Provide as much support as possible to the casualty during extrication. Whenever possible, give essential first aid and immobilize the injuries before the casualty is moved.

When there is an immediate danger and you are alone and must move a casualty from a vehicle, proceed as follows:



1. If necessary, disentangle the person's feet from the vehicle and bring the feet toward the exit. Ease your forearm under the person's armpit on the exit side, extending your hand to support the chin.

2. Ease the person's head gently backward to rest on your shoulder while keeping the neck as rigid as possible.

3. Ease your other forearm under the armpit on the opposite side and hold the wrist of the casualty's arm which is nearest the exit.

4. Establish a firm footing and swing around with the person, keeping as much rigidity in the neck as possible. Drag the casualty from the vehicle to a safe distance with as little twisting as possible.




Chapter 3

Airway and Breathing Emergencies



- Hypoxia
- Effective and ineffective breathing
- Inhalation injuries
- Breathing emergencies caused by illness
- Asthma
- Severe allergic reaction
- Choking

Chapter 3 Airway and breathing emergencies

When a person's breathing is affected through injury or illness, their life can be in immediate danger. As a first aider, you have to be able to recognize a breathing emergency very quickly and know what first aid to give—the casualty's life may depend on it.

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Hypoxia

Choking and breathing emergencies cause a lack of oxygen in the blood, a condition called hypoxia. This can damage vital tissues and eventually cause death. The causes of hypoxia are grouped under three headings:

1. **Lack of oxygen** —for example:
    - The oxygen level is low, such as at a high altitude
    - The oxygen is displaced by other gases, such as a build-up of silo gas in a grain silo
    - The oxygen in a small space is used up—for instance in a confined space
2. **Blocked airway**—for example:
    - A casualty chokes on a foreign object, such as food
    - An unconscious casualty's airway is blocked by their tongue
    - A casualty's airway becomes swollen due to an allergic reaction
3. **Abnormal heart and lung function**—where the heart and lungs are not working properly due to:
    - An illness such as pneumonia or congestive heart failure
    - An injury preventing effective breathing
    - A drug overdose or poisoning

Effective and ineffective breathing

The normal breathing rate varies for infants, children and adults. A breathing rate that is too slow or too fast is a sign of a breathing emergency. Breathing rhythm refers to the interval between breaths. In normal breathing, the intervals are even and breathing is effortless—this is regular breathing. Breathing depth refers to the amount of air moved in and out of the lungs with each breath. Signs of **effective breathing** include:

- Breathing that is quiet
- Equal expansion of both sides of the chest when the person inhales
- The person is alert and relaxed
- Skin colour is normal
- Speaking without taking a breath every few words

When a person is not getting enough oxygen, the body responds by breathing faster and deeper. Signs of **ineffective breathing** include:

- The casualty is struggling for breath or gasping for air
- Breathing rate is too fast or too slow
- Breathing rhythm is irregular
- Breathing depth is too shallow or too deep
- Breathing is noisy or raspy
- The person is "getting tired" from trying to breathe
- The person is sweating
- Decreased level of consciousness
- The lips, ears and fingernail beds turn blue—called cyanosis
- Chest movement may be abnormal

First aid for ineffective breathing

Always send or go for medical help at the first sign of a breathing emergency.

The first aid for ineffective breathing has two parts:

1. Give first aid for the injury or condition and position the responsive casualty in the semi-sitting position if possible
2. If breathing stops the casualty will become unresponsive, get medical help immediately and begin CPR

This table lists some of the causes of breathing emergencies. To give first aid, first determine the cause of the breathing emergency, and then decide on the best first aid actions.

Causes of airway and breathing emergencies		
Injuries	**Medical conditions**	**Poisoning**
Broken ribs	Asthma	Inhaled poison – e.g. carbon monoxide or hydrogen sulfide
Near drowning	Stroke	Swallowed poison – E.g. household cleaners or medication overdose
Knife or gunshot wound	Allergic reaction	Injected poison – e.g. bee sting
Burns to the face or airway	Pneumonia	
Head injury	Congestive heart failure	
Compression of the chest preventing chest expansion	Emphysema/bronchitis	

Inhalation injuries

Inhalation injuries happen when the casualty inhales hot steam or hot (superheated) air, smoke or poisonous chemicals.

Signs and symptoms of inhalation injuries include signs of shock:

- Dizziness, restlessness, confusion
- Pallor or cyanosis
- Abnormal breathing rate or depth

Together with a history of fire and:

- Noisy breathing
- Pain during breathing
- Burns on the face, especially the mouth and nose
- Singed hair on the face or head
- Sooty or smoky smell on breath
- Sore throat, hoarseness, barking cough, difficulty swallowing

First aid for an inhalation injury

1. Perform a scene survey and do a primary survey. Give first aid for the ABCs.
2. Place a conscious casualty in the semi-sitting position and loosen tight clothing at the neck, chest and waist.
3. If breathing stops, begin CPR starting with compressions.
4. Give ongoing casualty care until handover to medical help.

Breathing emergencies caused by illness

Illnesses that can lead to severe breathing difficulties include asthma, allergies, chronic obstructive pulmonary disease (e.g. emphysema), congestive heart failure, stroke and pneumonia.

Chronic Obstructive Pulmonary Disease (COPD)

Chronic Obstructive Pulmonary Disease is a term used to describe a group of respiratory conditions such as chronic bronchitis and emphysema. Casualties present with on-going shortness of breath and appear to be struggling to breathe. Some people may use supplemental oxygen delivered by nasal prongs from a small canister they carry when they have a more serious case of COPD.

Asthma

Asthma is a reactive airway illness in which the person has repeated shortness of breath, characterized by wheezing and coughing. A mild asthma attack is not a health emergency and can be managed by the casualty. A severe asthma attack can be fatal and requires immediate first aid. In response to a 'trigger' the person's airway can spasm, swell and secrete thick mucus, which narrows the airway passage.

Some common triggers that can cause asthma are:

- Colds, upper airway infections
- Pet dander
- Insect bites, stings
- Foods
- Pollen, paint and smoke

Signs and symptoms of a severe asthmatic attack:

- Shortness of breath with obvious trouble breathing
- Coughing or wheezing
- Fast, shallow breathing
- Casualty sitting upright trying to breathe
- Bluish colour in the face (cyanosis)
- Anxiety, tightness in the chest
- Fast pulse rate, shock
- Restlessness at first, and then fatigue

First aid for a severe asthma attack

1. Perform a scene survey and a primary survey; send for medical help.
2. Place the casualty in the most comfortable position for breathing. This is usually sitting upright with arms resting on a table.
3. Help the casualty take prescribed medication.
4. Give ongoing casualty care.
5. If the unconscious casualty stops breathing, begin CPR.

A person with asthma may carry medication in the form of a:

- Metered-dose inhaler (MDI)
- Turbuhaler
- Diskus®

Usually the person can give themselves this medication without help. If the person needs help, a first aider can assist.

An inhaler delivers a pre-measured amount of medication. Always read and follow the manufacturer's instructions. Check the prescription label to confirm the casualty's name and expiry date.

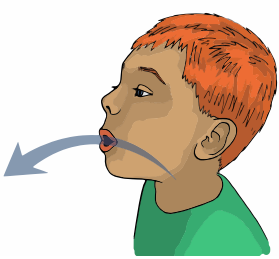
To assist with a Metered Dose Inhaler

The metered dose inhaler (or "puffer") is the more common method of delivering medication for asthma.



Check your applicable workplace legislation/ regulations

1. Shake the container, then remove the cap.
2. Tell the casualty to breathe out completely,



3. Tell the casualty to breathe in slowly and deeply—as the casualty does, the MDI will be pressed to release the medication. The MDI can be in the mouth, or just in front of the mouth.



4. Tell the casualty to hold their breath for 10 seconds so the medication can spread out in the lungs. Then tell them to breathe normally, so the medication won't be expelled. If more doses are needed, wait at least 30-60 seconds before repeating these steps.

Spacers (Aerochamber®)

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When the medication comes out of the inhaler, it may be deposited on the back of the throat and not reach the lungs or the casualty may be gasping for air and unable to hold their breath. To deal with this, use a spacer if available. It traps the particles of the spray, allowing the casualty to inhale more effectively over several breaths.

Spacers make it simple to inhale the medication, and should always be used when available. Small children and other casualties who have difficulties coordinating proper inhalation with the release of the medication will often have spacers with them.



It allows them to inhale two or three times before the medication is completely dispelled. A mask can be attached to the device to make taking the medication easier.



If the casualty complains of throat irritation after using the inhaler, have them gargle or rinse the mouth with water.

Severe allergic reaction

An allergic reaction occurs when the immune system reacts to a substance the body encounters. Most allergies are annoying but not dangerous.

Anaphylaxis is a severe allergic reaction which usually happens when a substance to which the casualty is very sensitive enters the body, although it can also be caused by exercise or have no known cause. Anaphylaxis can happen within seconds, minutes or hours of a substance entering the body. As a rule, the sooner the casualty's body reacts, the worse the reaction will be. Anaphylaxis is a serious medical emergency that needs urgent medical attention.

Common early **signs and symptoms** of an allergy may include itchy flushed skin with hives; sneezing and a runny nose; coughing. If it's a severe reaction there may be swelling of the face and neck, especially the lips and tongue. Breathing may become difficult if the swelling is internal too. There may be nausea and vomiting and the casualty may be anxious and feeling sense of impending doom as their blood pressure drops and they go into shock. This is a true medical emergency and requires immediate first aid.

First aid for a severe allergic reaction

1. Perform a scene survey and a primary survey. Send for medical help.
2. Place the casualty in the most comfortable position for breathing—usually sitting upright.
3. Assist the casualty with their medication, usually this is an epinephrine auto-injector.
4. Give ongoing casualty care.

It is important to be familiar with, and follow the manufacturer's instructions, which is located on the side of the auto-injector.

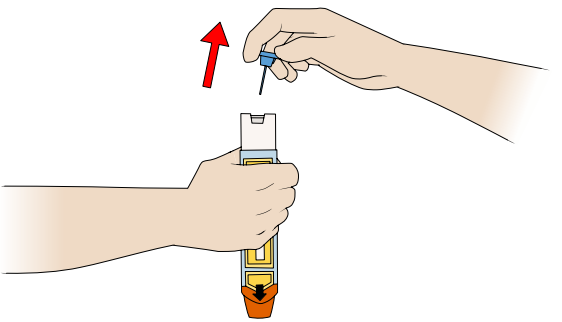
Check the expiry date. If the only auto-injector is an expired product, it may still save a life and should be administered anyway, if the indicated liquid remains clear.



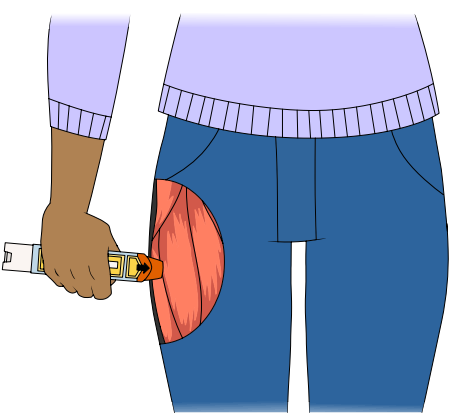
Check your applicable workplace legislation/ regulations

To use the auto-injector:

1. Remove the EpiPen® from the storage tube. Hold it firmly with the orange tip downward. Remove the blue safety release.



2. Use the auto-injector on only the fleshy part of the mid-outer thigh. Auto-injectors can be given through lightweight clothing.
3. Press the orange tip of the EpiPen® firmly into the mid-outer thigh until the unit activates.
4. Hold the auto-injector in place for several seconds, then pull it straight out.



5. After the injection, keep the casualty warm and avoid any exertion. Call 9-1-1 as soon as you have given the first dose.



If the casualty shows no improvement within 5 minutes or if their condition deteriorates before help arrives a second dose may be given if it is available. This will require a second EpiPen®. Individuals who are feeling faint or dizzy because of impending shock should be placed flat on their back unless they are vomiting or experiencing respiratory distress.

It is important that the casualty does not sit or stand immediately as this could cause a drop in blood pressure. The medication will begin to wear off within 10 to 20 minutes—get medical help right away.

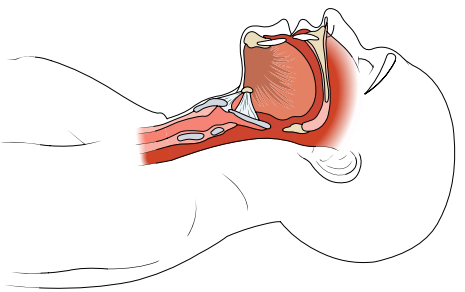
If you or anyone else is injected by mistake, get medical help.

Follow manufacturer's directions for proper care of the used device. Put the used unit back in the storage container and take it to the hospital with the casualty.

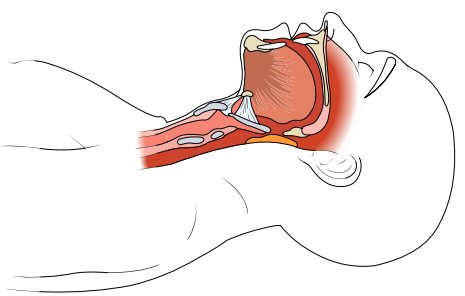
Choking

A person chokes when the airway is partly or completely blocked and airflow to the lungs is reduced or cut off. The choking casualty either has trouble breathing or cannot breathe at all.

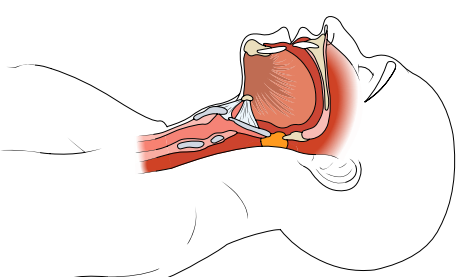
Open and clear airway



Partly blocked airway



Completely blocked airway



Causes of choking		
Foreign objects	**The tongue**	**Swelling**
Infants and children—food, toys, buttons, coins, etc.	Tongue falls to the back of the throat when lying on back	Injury to the throat area causes swelling of the airway
Adults—consuming drinks quickly with food in your mouth	Saliva, blood or vomit pools in the throat	Illness causes swelling, e.g. allergic reaction, asthma, epiglottis, croup
In elderly people—food, pills		Swollen airway

With good air exchange, the obstruction is mild and person can still cough forcefully, breathe and speak. With poor air exchange, the obstruction is severe and the person cannot cough forcefully, has trouble breathing, or cannot speak. With a completely blocked airway, there is no air exchange—coughing, breathing and speaking are impossible.

When the air supply to the lungs is cut off, the person's face immediately becomes red or "flushed". Shortly after, as the oxygen in the body is used up, the face becomes grey and the lips and ear lobes become blue. The person then becomes unconscious and eventually the heart stops beating.

Signs of choking	
Mild obstruction	**Severe obstruction**
Able to speak	Not able to speak
Signs of distress—eyes show fear	Signs of distress—eyes show fear
Forceful coughing	Weak or no coughing
Wheezing and gagging between coughs	High-pitched noise or no noise when trying to breathe
Red or "flushed" face	Grey face and blue lips and ears

First aid for choking

First aid for a choking adult or child

1. Perform a scene survey.
2. If the casualty can cough forcefully, speak or breathe, tell them to try to cough up the object. If a mild obstruction lasts for a few minutes, get medical help.

If you think there might be a severe obstruction, check by asking, "Are you choking?" If the casualty cannot cough forcefully, speak or breathe, use back blows followed by abdominal thrusts to remove the blockage.

1. Give back blows and abdominal thrusts:
2. Support the casualty and give up to five blows between the shoulder blades using the heel of your hand.
3. If the obstruction is not cleared, step behind the casualty ready to support them if they become unconscious.
4. Make a fist, place it on the casualty's abdomen at the belly button, in line with the hip bones. Grasp the fist with the other hand and give five



forceful inward and upward abdominal thrusts.

5. If the object is not removed, repeat back blows and abdominal thrusts.

If the casualty becomes unconscious



1. Lower them to the ground. Call for medical help and get an AED if available.
2. Begin chest compressions immediately. After the first 30 compressions, check the mouth. Remove any foreign object you can see. Try to give 2 breaths. If air does not go in, continue to give chest compressions and inspecting the mouth before ventilations.

First aid for a choking casualty much larger than the rescuer

If a choking casualty is very large or is in the late stages of pregnancy, give back blows as normal, followed by chest thrusts.



1. While supporting the casualty, give up to five back blows between the shoulder blades, using the heel of your hand.
2. If the obstruction is not cleared, stand behind the casualty.
3. Keep your arms horizontal and snug up under their armpits.
4. Place your fist against the lower half of the breastbone, thumb-side in.
5. Hold your fist with your other hand. Pull inward forcefully.
6. Continue giving back blows and chest thrusts until either the object is removed or the casualty becomes unconscious.



Choking adult – self-help

If you begin to choke on an object you may have to clear your own airway.

1. If there are people around, get their attention, do not isolate yourself from others.
2. Try to cough up the object.
3. Give yourself abdominal thrusts until you can cough forcefully, breathe or speak.

A second method is to use a solid object like the back of a chair, a table or the edge of a counter.

- Position yourself so the object is just above your hips.
- Press forcefully to produce an abdominal thrust.



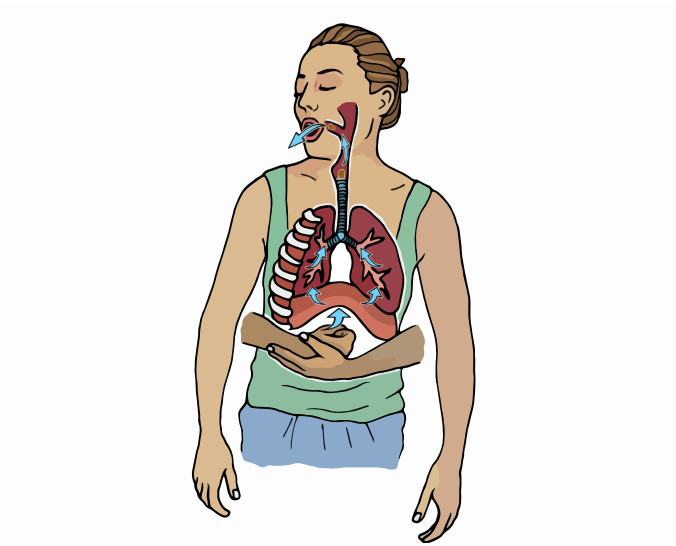
How abdominal thrusts work

When you choke on something, your body tries to unblock your airway by coughing.

Abdominal thrusts try to do the same thing with an artificial cough. The illustration below shows how an abdominal thrust creates a cough.

An abdominal thrust pushes the diaphragm up towards the lungs very quickly. This forces the air from the lungs up the airway and hopefully blows the obstruction out.

For the best effect, the fist has to be in the correct place. Keep your forearms off the abdomen and make each thrust a strong and sudden movement.



First aid for a choking casualty in a wheelchair

If you can reach around from behind the wheelchair, give back blows as normal, and abdominal or chest thrusts. If you cannot reach around the wheelchair:

- Position the wheelchair against a wall.
- Put the wheelchair brake on.



- If possible, carefully lean the casualty forward and support the shoulders. Perform five back blows between the shoulder blades, using the heel of your hand.
- Put the heel of one hand, with the other on top, in the middle of the abdomen and give up to five sudden, inward/upward thrusts.
- Alternatively, place the heel of one hand on the center of the breastbone and give firm chest thrusts.
- Repeat back blows and abdominal or chest thrusts until the object is removed or the casualty becomes unconscious.

If a doctor, physiotherapist or other health professional has shown you a different way of giving abdominal thrusts to a person in your care, use the recommended method.

If the casualty becomes unconscious, take them out of the wheelchair.

- Call for medical help and get an AED.



- Pull the casualty forward supporting them as best as you can and lower them to the ground.
- Roll the casualty to the floor to a face-up position.



- Begin chest compressions immediately. After the first 30 compressions, check the mouth. Remove any foreign object you can see. Try to give 2 breaths and continue to give chest compressions and inspecting the mouth before ventilations.

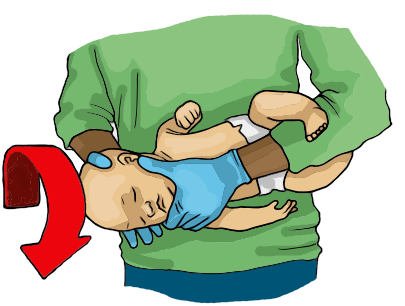
First aid for a choking infant

An infant is choking when they suddenly have trouble breathing, coughing, gagging, with high-pitched, noisy breathing.

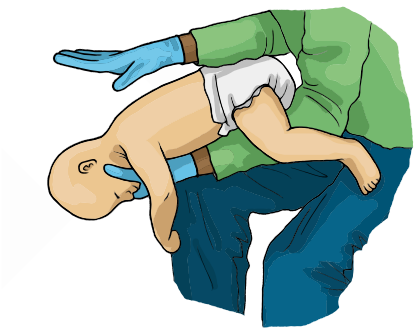
1. Perform a scene survey and primary survey.
2. If the baby can cough forcefully or breathe let the baby try to cough up the object. If a mild obstruction lasts for more than a few minutes, send for medical help.



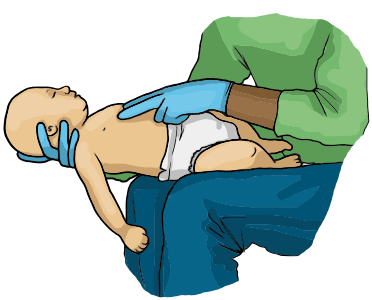
3. If the baby cannot cough forcefully, cannot breathe, makes a high-pitched noise when trying to breathe or starts to turn blue, begin back blows and chest thrusts.
4. Secure the baby between your forearms and turn them face down.



5. With the baby's head lower than the body, use the heel of your hand to give five forceful back blows between the shoulder blades.



6. Turn the baby face-up and give five chest thrusts.



7. Keep giving back blows and chest thrusts until either the airway is cleared or the baby becomes unconscious.
8. If the baby becomes unconscious, send for medical help. Begin chest compressions immediately.
After the first 30 compressions, check the mouth. Remove any foreign object you can see. Try to give 2 breaths and continue to give chest compressions and inspecting the mouth before ventilations



Chapter 4

Cardiovascular Emergencies and Cardiopulmonary Resuscitation (CPR)



- Cardiovascular disease
- Chain of survival ®
- Angina and heart attack
- Stroke and transient ischemic attack (TIA)
- Cardiac arrest
- Cardiopulmonary resuscitation (CPR)
- Automated external defibrillation - (AED)

Chapter 4 Cardiovascular emergencies and Cardiopulmonary Resuscitation (CPR)

Cardiovascular disease

Cardiovascular disease is one of the leading causes of death of adults in Canada. Some of these deaths could be prevented if appropriate first aid was given. This chapter describes the first aid for cardiovascular emergencies, including

- First aid for angina/heart attack
- First aid for stroke/TIA
- First aid for cardiac arrest, which is CPR

High blood pressure



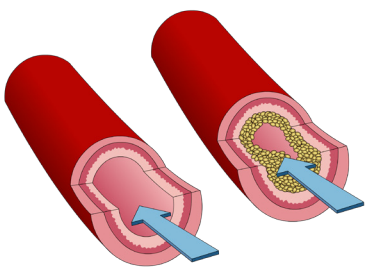
Blood pressure is the pressure of the blood against the inside walls of the blood vessels. Blood pressure goes up and down naturally. When a person is excited or emotionally stressed, blood pressure goes up, but it usually comes down once the excitement has passed. In some people, their blood pressure stays high all the time. This condition of constant high blood pressure is called hypertension.

Over time, hypertension damages the tissues of the cardiovascular system. The walls of the blood vessels become thick and lose their elasticity and the heart becomes enlarged. The changes caused by high blood pressure increase the risk of stroke, heart attack, kidney, and eye problems. Unfortunately, hypertension does not always give warning signals—you may feel perfectly well but still have high blood pressure. This is why it is often called the "silent killer."

If you have concerns about your blood pressure, you should speak with your primary health-care provider. Blood pressure machines located in pharmacies can help you monitor your blood pressure, but should never be used as a means of self-diagnosis.

Narrowing of the arteries

Arteries are the blood vessels that carry blood away from the heart. They become diseased when fatty deposits build up inside them, making the passage for blood narrower. This process of depositing fat and narrowing of the arteries is called atherosclerosis. In the coronary arteries, which carry oxygenated blood to the heart, it is called coronary artery disease.



As an artery gets narrower, less and less blood can get through. When the artery gets too narrow, the tissues on the other side of the narrowing don't get enough oxygenated blood to function normally. Although the signs and symptoms of hardening of the arteries usually don't appear until middle age or later, atherosclerosis often begins in childhood.

Angina

Angina occurs when the blood supply feeding the heart muscle becomes limited due to narrowed, damaged, or blocked arteries. When the heart works harder and needs more blood (e.g. when you run for a bus or shovel snow), it cannot get enough blood. This causes pain or discomfort in the chest, which may spread to the neck, jaw, shoulders, and arms. Angina pain typically doesn't last long, and goes away if the person rests and takes their prescribed medication.

Heart attack

A heart attack happens when heart muscle tissue dies because its supply of blood has been cut off. A heart attack can feel just like angina, except the pain doesn't go away with rest and medication. If the heart attack damages the heart's electrical system, or if a lot of the heart muscle is affected, the heart may stop beating properly. This is cardiac arrest.

Risk Factors

A number of factors increase the risks for cardiovascular disease, heart attack, and stroke. These can be broken down into modifiable and non-modifiable risks.

The modifiable risk factors can lead to dyslipidemia (increased deposits of fats), obesity, diabetes, and high blood pressure.

Non-modifiable	Modifiable
• Age • Genetic History • Sex	• Smoking • Poor diet • Lack of exercise • Increased stress

Modifiable risks can be reduced through lifestyle changes.

Angina and heart attack

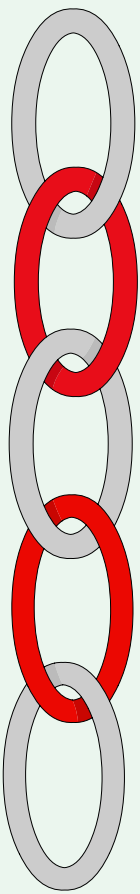
Early recognition and denial

The first step is recognizing a cardiovascular emergency. It's difficult to accept that someone is having a heart attack and could die very soon, especially if the person is a family member or a close friend. The casualty is often denying anything serious is happening as well, so it's easy to accept their reassurances.

On average, casualties take several hours to get to a hospital from the time they first start feeling poorly. It is this wasted time that prevents many lives from being saved. When someone complains of chest pain, shortness of breath and looks odd you should consider it a serious problem—that's early recognition, and call for medical help. Getting the casualty to the hospital quickly gives them the best chance for survival.

Chain of Survival®

CPR is often what comes to mind when people think of first aid for a heart attack or cardiac arrest. But CPR is only part of the picture. There are five steps that are important when helping someone with heart problems.



1. Immediate **recognition** of a cardiovascular emergency and activation of the community emergency medical services (EMS) system. This means calling for help quickly.
2. Early **CPR** with an emphasis on chest compressions.
3. Rapid **defibrillation**.
4. Effective **advanced life support**.
5. Integrated **post-cardiac arrest care**.

Each of the steps is as important as the others. Time is a vital ingredient. To give a casualty in cardiac arrest a reasonable chance of survival, CPR must be started immediately followed by defibrillation as quickly as possible. For both procedures, the sooner they happen, the better.

You, the first trained person on the scene, are responsible for initiating the sequence. You must recognize the cardiovascular emergency, call for medical help, start CPR if needed, and apply a defibrillator if one is available. You are the crucial first three links in the Chain of Survival®.

If nothing is wrong the ambulance crew can reassure the casualty. On the other hand, if there is a serious problem, you may have saved a life.

Signs and symptoms of angina and a heart attack:

A heart attack will produce shock and may display some or all of the following:

- Pale, ashen skin
- Sweating, cold and clammy to the touch
- Shortness of breath
- Showing obvious pain or discomfort

The pain or discomfort will be in the upper body, from the upper abdomen to the jaw and arms, and may feel like:

- Heaviness in chest
- Tightness or pressure in chest
- Squeezing or crushing chest
- Indigestion, nausea or vomiting
- Aching jaw
- Sore shoulder or arms



Some other **signs and symptoms** include:

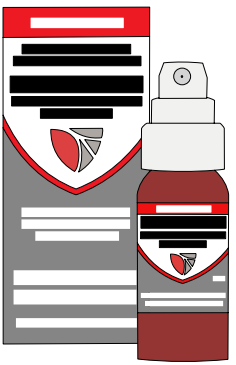
- Fatigue
- Anxiety, which produces denial
- Central back pain

Denial is an important detail. If someone showing signs of shock, having trouble breathing and experiencing pain insists there is nothing wrong, then you should be very suspicious and take action.

First aid for angina/heart attack

1. Perform a scene survey, then do a primary survey. Ask the casualty questions :
    - "Can you show me where it hurts?"
    - "Have you had this pain before?"
    - "Do you have medication for this pain?"
2. Call for medical help and get a defibrillator.
3. Place the casualty at rest, the semi-sitting position is usually the best option, and reassure them.
4. Assist the conscious casualty to take their prescribed medication, usually nitroglycerin. If the casualty has no prescribed medication, or the first dose is ineffective, ask the casualty if they have any allergies to ASA, or if a doctor has ever told them not to take it. If the casualty believes they can take it, suggest they chew one regular ASA tablet (or two low-dose tablets). ASA can reduce the effects of a heart attack because of its anti-clotting properties.
5. If the casualty loses consciousness and stops breathing, start CPR.

Helping with Nitroglycerin





Nitroglycerin tablets or sprays are common medications for relief of chronic angina pain. A casualty in serious distress may need your help to take their medication.

Ask the casualty if they have taken any other medications today. Drugs to treat erectile dysfunction such as Viagra® or CIALIS® may cause a significant decrease in the person's blood pressure if nitroglycerin is taken as well.

Have the casualty spray the medication under the tongue or place the tablets under the tongue—they aren't to be swallowed.

Nitroglycerin may be repeated, if needed, every 5-10 minutes to relieve pain, or until a maximum of three doses have been taken. Remember that if you have to assist someone to take their medication, you must call for medical help!

Stroke and transient ischemic attack (TIA)

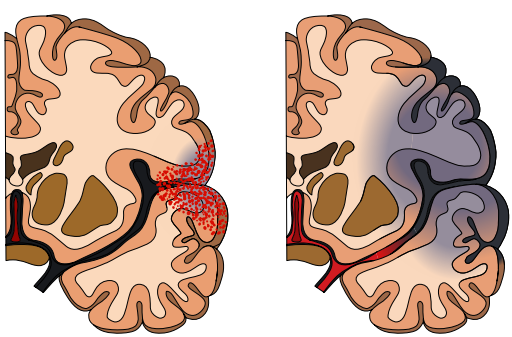
Stroke

A stroke happens when blood flow to a part of the brain is interrupted either by a blocked artery or by a ruptured blood vessel in the brain. A stroke may cause brain damage which impairs certain body functions, depending on the part of the brain affected.

Transient ischemic attack (TIA)

A TIA is a temporary blockage of the blood flow to part of the brain. It's typically of short duration and leaves no permanent damage but looks exactly like a stroke.

Doctors now have therapies to restore blood flow to the heart muscle and brain, but they work best if used right away. This is why it's important to realize there's an emergency and call 9-1-1 to get the casualty to the hospital right away—the longer medical help is delayed, the more likely the heart or the brain will be damaged.



Remember **FAST** as a way to check for the **signs and symptoms** of a stroke and to get immediate help.

- **Facial droop**. Ask them to smile. One side of the face may not move as well as the other side.
- **Arm drift**. Ask the casualty to hold both arms out with the palms up, and close their eyes. One arm may not move or drifts down compared to the other arm.
- **Speech**. Ask them to repeat a phrase you say. The casualty may slur words, use the incorrect words or is not able to speak.
- **Time**. When was the onset of symptoms? Ask the casualty, or their family, friends, or bystanders when the symptoms were first noticed. Get immediate medical help; the earlier a stroke is treated the better the outcome.

Other signs and symptoms of a stroke include

- Blurred vision
- Sudden confusion
- Dizziness
- Headache
- Loss of balance.

It is important that first aid providers do not dismiss the signs and symptoms of a stroke as intoxication.

First aid for stroke/TIA

1. Perform a scene survey, then do a primary survey; perform the FAST assessment.
2. Call for medical help.
3. Place the casualty at rest in the semi-sitting position.
4. Give nothing by mouth, especially ASA.
5. Give ongoing care.

If the casualty becomes unconscious, place them in the recovery position. If there is paralysis, position the casualty with the paralyzed side up. This will reduce the chance of tissue or nerve damage to the affected side.

Cardiac arrest

Cardiac arrest means the heart stops beating properly. With no blood flow going to the brain the casualty becomes unresponsive and stops breathing. Cardiac arrest means the casualty is clinically dead, but if CPR is started and a defibrillator is applied quickly there is still an opportunity to restore a normal heartbeat.

Common causes of cardiac arrest include:

- Heart attack
- Severe injuries
- Electrical shock
- Drug overdose
- Drowning
- Suffocation

Cardiopulmonary Resuscitation (CPR)

CPR is artificial respiration and artificial circulation. Artificial respiration provides oxygen to the lungs. Artificial circulation causes blood to flow through the body. The purpose of CPR is to circulate enough oxygenated blood to the brain and other organs to delay damage until either the heart starts beating again, or medical help takes over from you. CPR is most effective when interruptions to chest compressions are minimized.

CPR – Adult casualty

1. Perform a scene survey.
2. Assess responsiveness.
3. If there is no response, call for medical help on a mobile device, and place the phone on speaker-phone, and send someone for an AED. If no mobile phone is available, send or go for medical help and the AED, if available.

4. Perform a primary survey:
    - Open the airway.
    - Check for breathing for at least 5 and no more than 10 seconds.
5. If the casualty is not breathing, or not breathing effectively



(agonal breaths) position your hands in the centre of the upper chest and your shoulders directly over your hands. Keep your elbows locked.



6. Give 30 compressions—Push hard—Push Fast!
    - Press the heels of the hands straight down on the breastbone. The depth of each compression should be at 5-6 cm (2-2.4 inches).
    - Release pressure and completely remove your weight at the top of each compression to allow chest to return to the resting position.
    - Give compressions at a rate of 100 to 120 per minute. Count compressions out loud to keep track of how many you have given, and to help keep a steady rhythm.

7. Open the airway by tilting the head and lifting the chin.
8. Position a barrier device and breathe into the casualty twice. For an adult casualty, each breath should take about for 1 second, with just enough air to make the chest rise.



This is one cycle of 30:2 (30 compressions to 2 ventilations).

9. Continue CPR until either an AED is applied, the casualty begins to respond, another first aider or medical help takes over or you are too exhausted to continue. The AED should be applied as soon as it arrives at the scene.

Agonal breathing

Agonal breathing is an abnormal pattern of breathing driven by a brain-stem reflex, characterized by irregular gasping respirations at times accompanied by strange vocalizations. They can occur with cardiac arrest and lead bystanders to believe the casualty is breathing. A casualty with agonal breathing should be treated as though they are not breathing.

CPR – Child casualty

1. Perform a scene survey.
2. Assess responsiveness.
3. If there is no response, send or call for medical help and an AED if available.

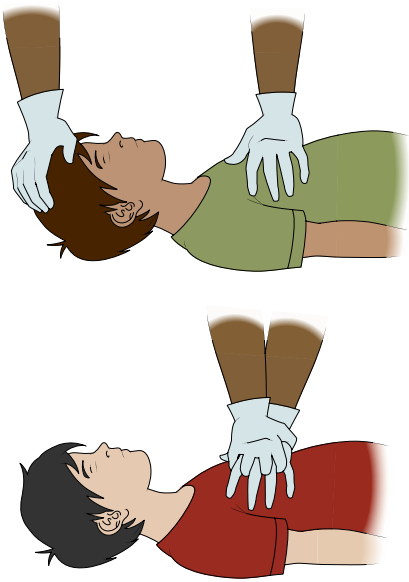
If you are alone with no phone perform 5 cycles of CPR (two minutes) then go for medical help. Carry the child with you if possible.

4. Perform a primary survey:
   - Open the airway.
   - Check for breathing for at least 5 and no more than 10 seconds.



5. If the casualty is not breathing, or not breathing effectively (agonal breaths) position your hands in the centre of the upper chest and your shoulders directly over your hands. Keep your elbows locked. You may use one or two hands depending on the size of the child.

6. Give 30 compressions—Push hard—Push Fast!



- Press the heels of the hands straight down on the breastbone. The depth of each compression should be 1/3 of the chest depth, or 5 cm (2 inches).
- Release pressure and completely remove your weight at the top of each compression to allow chest to return to the resting position.
- Give compressions at a rate of 100 to 120 per minute. Count compressions out loud to keep track of how many you have given, and to help keep a steady rhythm.

7. Open the airway by tilting the head and lifting the chin.
8. Position a barrier device and breathe into the casualty twice, with just enough air to make the chest rise.

This is one cycle of 30:2 (30 compressions to 2 ventilations).

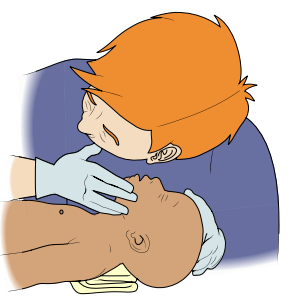
9. Continue CPR until either an AED is applied, the casualty begins to respond, another first aider or medical help takes over or you are too exhausted to continue. The AED should be applied as soon as it arrives to the scene.

CPR – Infant casualty

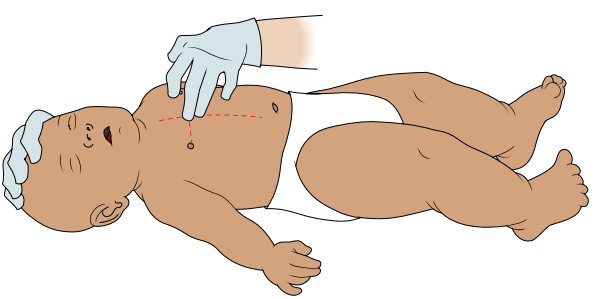
1. Perform a scene survey.
2. Assess responsiveness. Gently tap the baby's feet.
3. If there is no response, send or call for medical help and an AED if available.

If you are alone with no phone perform 5 cycles of CPR (two minutes) then go for medical help. Carry the infant with you if possible.

4. Perform a Primary Survey
   - Open the airway.
   - Check for breathing for at least 5 and no more 10 seconds.



5. If the baby is not breathing, or not breathing effectively (agonal breaths) begin CPR



- Place two fingers on the breastbone just below the nipple line. Push down on the breastbone 1/3 the depth of the chest or about 4 cm (1 1/2 inches).

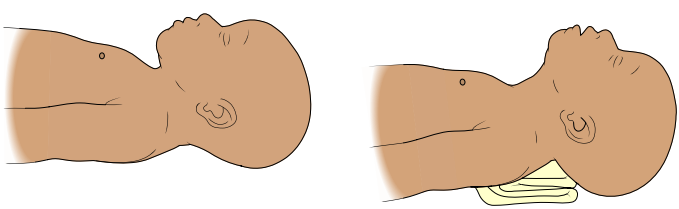
- Release the pressure completely but keep your fingers in light contact with the chest. Repeat the pressure and release phases rhythmically so that each phase takes the same amount of time.
- Give compressions at a rate of 100 to 120 per minute. Count compressions out loud to keep track of how many you have given, and to help keep a steady rhythm.

6. Open the airway by tilting the head and lifting the chin.
7. Position a barrier device and breathe into the casualty twice, with just enough air to make the chest rise.

This is one cycle of 30:2 (30 compressions to 2 ventilations).

8. Continue CPR until either an AED is applied, the casualty begins to respond, another first aider or medical help takes over or you are too exhausted to continue. The AED should be applied as soon as it arrives to the scene.

The back of an infant's head is quite large compared to the rest of the body. This causes the baby's head to come forward and close off their airway.



An infant's head flexes forward when they are lying on their back.

When giving CPR, it may be helpful to put a thin pad under the shoulders to help keep the airway open—but don't waste time looking for a pad.

Chest compression only CPR

CPR guidelines stress early recognition of the emergency and stress the importance of calling 9-1-1 or the local emergency number if you find someone collapsed and unresponsive.

If you have not been trained in CPR or are hesitant to perform ventilations, for any reason—don't give up. Your actions can still save a life.

Compression only CPR is CPR without mouth-to-mouth breaths. Provide high quality chest compressions by pushing hard and fast on the centre of the chest, at a rate of 100 to 120 compressions per minute.

Although this does not give the casualty any oxygen, this option can be used by people not trained in conventional CPR, or those who are unsure of their ability.



Dispatcher-assisted CPR

In many locales, the 9-1-1 dispatcher is trained to coach you through an emergency until medical help arrives. Put your phone on speaker and place it by the casualty's head and talk to the dispatcher throughout the rescue.



How to take over CPR from another rescuer

1. Offer to help, tell the rescuer that you are trained in CPR. Ensure medical help has been called.



2. Give 30 compressions followed by 2 breaths. Use your own barrier device if available





Two-rescuer CPR

If two trained rescuers are available, they can cooperate to perform CPR on a casualty. There are three advantages to two rescuers performing CPR as a team:

- CPR is a strenuous physical activity and as a first aider gets tired the quality of the chest compressions will deteriorate. By sharing the task of compressing the chest two rescuer CPR allows for a team to perform effective chest compressions for a longer period of time.
- Two-rescuer CPR minimizes the time the compressions are interrupted for ventilations to be given.
- Two-rescuer CPR allows the rescuers to give feedback and support each other during a stressful event.

To perform two-rescuer CPR the first aider who performs the primary survey stays at the casualty's head, keeping the airway open and ventilating after 30 compressions. The second rescuer will compress the chest, but in order to maintain the most effective compressions, it is recommended that rescuers switch after every 5 cycles of compressions and ventilations (approximately 2 minutes).

Automated External Defibrillation—AED

Automated external defibrillation, the application of an electric shock to a heart that has stopped pumping effectively, has been proven to be one of the most important tools in saving the lives of sudden cardiac arrest casualties. It is the third link in the Chain of Survival® and is the responsibility of the first aider.

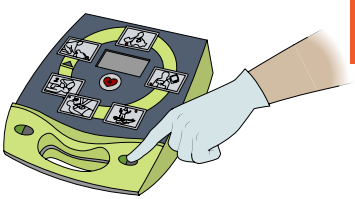
An automated external defibrillator (AED) is an electronic device that is programmed to recognize and shock two types of heart rhythms, Ventricular Fibrillation (VF) and pulseless Ventricular Tachycardia (VT). If the machine recognizes either VT or VF in a casualty, it will charge and will indicate that a shock is advised. The purpose of this shock is to correct the abnormal electrical disturbance and re-establish the heart rhythm.

It is important to remember that AEDs will only shock when VT or VF is present. You cannot shock a heart that is in normal rhythm, nor will the machine shock when it is not appropriate, such as when the heart is stopping (asystole) or there is pulseless electrical activity (PEA)

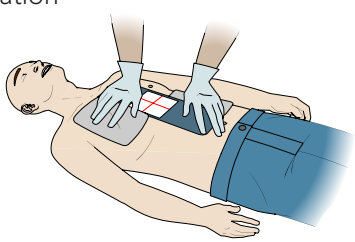
Time is a critical factor in determining survival from cardiac arrest; the heart will only stay in fibrillation a short time before all electrical activity ceases. Defibrillation must be performed early to be most effective. CPR can keep oxygenated blood flowing to the brain, and helps extend the length of time that the heart will remain in VT or VF, the only arrhythmias that AEDs will shock. CPR then can "buy some time" for the casualty until the AED is attached and ready to deliver a shock.

Using an AED (always follow the AED's voice prompts)

1. Power on the AED.
2. Follow the voice prompts. The audio instructions will direct you to:

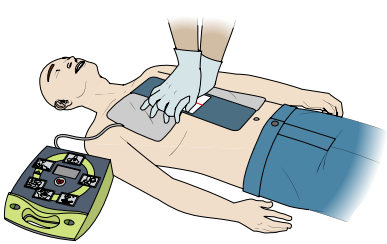


   - Bare the chest and attach electrode pads. The pads need to stick directly to the skin, so excessive sweat, water, and chest hair needs to be removed before application



   - Stand back (or clear)
   - Press the shock button and/or continue CPR as prompted by the machine

Continue with CPR and listen for the AED to give additional instructions



Defibrillation—Special Considerations and Special Circumstances

Pregnant patients–AEDs can be used in all stages of pregnancy.

Pacemakers or implanted defibrillators–Defibrillator pads should not be placed directly over a pacemaker site but should be approximately 2.5 cm (one inch) away. Look for scars or lumps on the chest as an indicator of implanted devices.

Children under 8 years of age–Automated external defibrillators (AEDs) may be used for children and infants. Special pads or a pediatric setting on the machine are used, but if not available adult pads can be used. Some adult pads show an alternate placement for children/infants.

Patch medications–some casualties wear a patch that contains medication such as nitroglycerin for angina. If the patch is in the way of the pad placement, gently remove it with gloved hands from the chest and wipe the area clean.

Wet environment–AEDs can be used in wet areas. Dry the chest to ensure good pad contact. Move the casualty to a dry area if possible. If you or the casualty is submersed in water, avoid using the AED.

Metal surfaces–AEDs can be used safely with the casualty on a metal surface.

Jewelry and piercings–Avoid placing pads over-top of piercings, jewelry, or anything that would cause a gap. AED pads should adhere flat to the skin.

Environment–Ensure the environment you are using an AED in does not contain explosive gases.

Post-resuscitation care and handover to EMS

If defibrillation is successful, the casualty may start breathing on their own but remain unresponsive. In this case, place the casualty into the recovery position and monitor the ABCs. Leave the AED attached. The AED will continually monitor the heart rhythm or you may need to use the device again.

Certain information is important for emergency services personnel such as the time of collapse, time when CPR was started, time when first shock was delivered and number of shocks. Provide as much detail as possible and follow the directions of medical personnel once they arrive on the scene.

Regulations concerning the requirement of a workplace to have an AED, and the necessary policies about AEDs, will be contained within federal, provincial, or territorial legislation. Where not specifically outlined by regulations, a workplace should have an established AED policy which outlines:

- Certification and recertification requirements
- Maintenance and inspection processes
- Post-use process (downloading information, resupplying pads and rescue pouch, etc.)
- Replacement of batteries and pads
- Replacement of unit at its end-of-life

Troubleshooting and maintenance

Sometimes the device will indicate "Check Electrodes". If this occurs, check the cable to pads connection, the cable to machine connection and the adherence of the pads to the casualty's chest.

Machines will also advise if motion is detected or if the battery is low. AEDs are sold with an instruction manual that will outline troubleshooting in detail.

Regular maintenance of AED units, including regular inspections, are important to ensuring the AED is available when it is needed. Most AEDs perform a daily system check and display in some manner it is ready for use. A monthly check of the unit is recommended to ensure the pads are still good (they have an expiry date), the rescue pack is present and stocked, and the status is green. Always follow manufacturer suggested guidelines and checklists.



Chapter 5

Wounds and Bleeding



- Dressings, bandages and slings
- Wounds and bleeding
- Bleeding
- Recognizing internal bleeding
- Chest injuries
- Abdominal injuries
- Crush injuries
- Head and facial injuries
- Eye injuries
- Burns
- Bites and stings

Chapter 5 Wounds and bleeding

Dressings, bandages, and slings

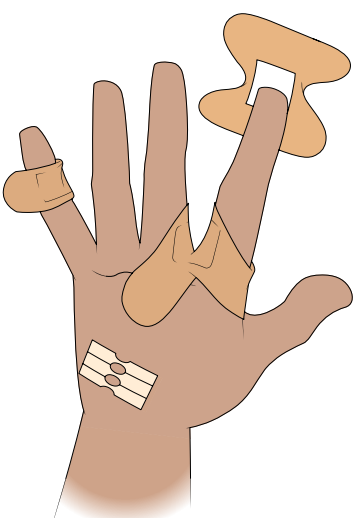
Dressings

A dressing is a protective covering put on a wound to help control bleeding, absorb blood from the wound, and prevent further contamination. A dressing should be:

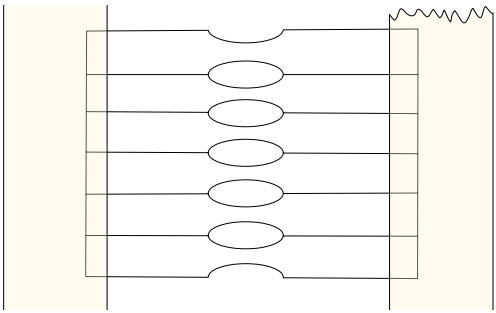
- Sterile, or as clean as possible
- Large enough to cover the wound
- Highly absorbent
- Compressible, thick and soft
- Non-stick and lint-free to reduce the possibility of sticking to the wound

Dressings are available in a variety of sizes and designs. The dressings used most often in first aid are:

- **Adhesive dressings** – prepared sterile gauze dressings with their own adhesive strips



- **Wound closures** – adhesive strips that bring the edges of the wound together to assist healing.



- **Gauze dressings** – packaged gauze available as sterile single packs or in bulk packaging
- **Pressure dressings** – large sterile dressings of gauze and other absorbent material, usually with an attached roller bandage. They are used to apply pressure to a wound with severe bleeding
- **Improvised dressings** – prepared from lint-free sterile or clean absorbent material such as a sanitary pad
- **Hemostatic dressings** – pressure dressings impregnated with clot promoting agents used to stop serious bleeding. These dressings are not designed for all wound types. Check with your local protocols for more information.

Follow the **guidelines** below for putting on dressings:

- Prevent further contamination
- Extend the dressing beyond the edges of the wound
- If blood soaks through a dressing, leave it in place and cover with more dressings
- Secure a dressing with tape or bandages

Bandages

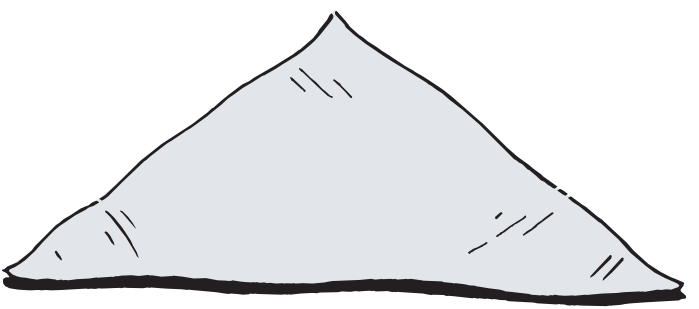
A bandage is any material that is used to hold a dressing in place, maintain pressure over a wound, support a limb or joint, immobilize parts of the body or secure a splint.

When using bandages, remember to:

- Apply firmly to make sure bleeding is controlled or immobilization is achieved
- Check the circulation below the injury before and after applying a bandage, you may have applied it too tightly or swelling may have made it too tight.

5

The Triangular Bandage



A triangular bandage may be used:

- As a whole cloth—opened to its fullest extent, as a sling or to hold a large dressing in place
- As a broad bandage—to hold splints in place or to apply pressure evenly over a large area
- As a narrow bandage—to secure dressings or splints or to immobilize ankles and feet in a figure-8

Broad Bandage

To form a broad bandage, fold the point to the centre of the base with the point slightly beyond the base

Fold in half again from the top to the base.



Narrow Bandage



Fold a broad bandage in half again from the top to the base to form a narrow bandage.

Reef knot—the knot of choice

The reef knot is the best knot for tying bandages and slings:

- It lies flat, making it more comfortable than other knots
- It does not slip
- It is easy to untie in order to adjust the bandage

To tie a reef knot:

- Take one end of a bandage in each hand
- Lay the end from the right hand over the one from the left hand and pass it under to form a half-knot. This will transfer the ends from one hand to the other
- The end now in the left hand should be laid over the one from the right and passed under to form another half-knot. The finished knot looks like two intertwined loops
- Tighten by pulling one loop against the other or by pulling only on the ends

Place knots so they do not cause discomfort by pressing on skin or bone, particularly at the site of a fracture or at the neck, when tying a sling.



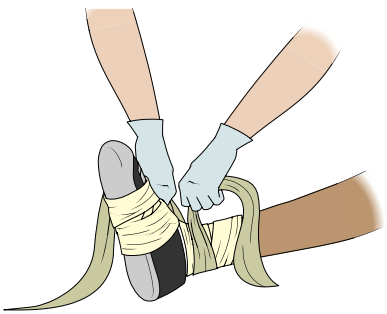
If the knot is uncomfortable, place soft material underneath as padding.

Figure-8

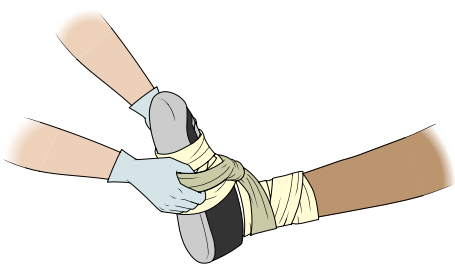
A figure-8 tie may be used to tie the ankles and feet, to secure a splint to the ankles/feet, or to support an injured ankle.

To tie a figure-8:

- Position the centre of a narrow or broad triangular bandage under the ankle (or both ankles if tying the feet together).



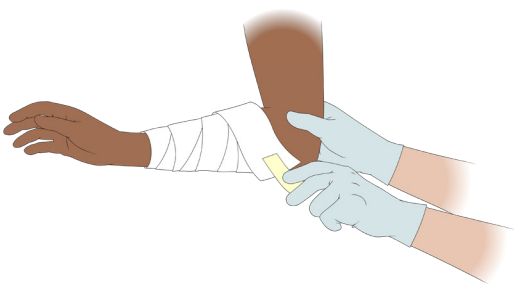
- Cross the ends over top the ankles, and bring the ends around the feet and tie off.



Roller bandage

Roller bandages, usually made of gauze-like elastic material, are used to hold dressings in place or to secure splints.

Put on a roller bandage in a simple spiral. Starting at the narrow part of the limb, anchor the bandage with a few turns and continue wrapping the bandage, overlapping each turn by one quarter to one third of the bandage's width. Make full-width overlaps with the final two or three turns and secure with a safety pin, adhesive tape or by cutting and tying the bandage as shown. Always check circulation below the wound before and after applying a bandage, you may have applied it too tightly or swelling may have made it too tight.



Slings

A sling can be easily improvised with a scarf, belt, necktie or other item that can go around the casualty's neck. You can also support the arm by placing the hand inside a buttoned jacket or by pinning the sleeve of a shirt or jacket to the clothing in the proper position.

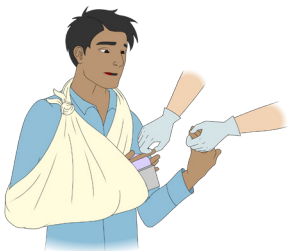
Arm sling

To put on an arm sling:

1. Support the forearm of the injured limb across the body. Place an open triangular bandage between the forearm and the chest so the point extends beyond the elbow and the base is straight up and down.



2. Bring the upper end around the back of the neck to the front of the injured side. While still supporting the forearm, bring the lower end of the bandage over the hand and forearm and tie off on the injured side in the hollow of the collarbone. Place padding under the knot for comfort.



3. Twist the point into a "pigtail" at the elbow and tuck it inside the sling.
4. Adjust the sling so you can see the fingernails—this way you can watch them to check on circulation.

St. John tubular sling

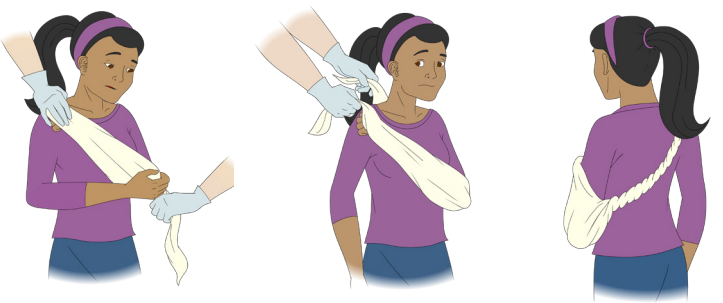
This sling is used for injuries to the shoulder or collarbone. To put on a St. John tubular sling:

1. Support the forearm of the injured side diagonally across the chest, the fingers pointing toward the opposite shoulder.
2. Place a triangular bandage over the forearm and hand with the point extending beyond the elbow and the upper end over the shoulder on the uninjured side. The base is placed vertically in line with the body on the uninjured side.



3. Ease the base of the bandage under the hand, forearm and elbow. Tuck the base of the bandage under the injured arm to make a pocket that runs the full length of the arm.

4. Gather the bandage at the elbow by twisting it and bring the lower end across the back and over the shoulder on the uninjured side. This closes the pocket at the elbow.
5. Gently adjust the height of the arm as you tie off the ends of the bandage so the knot rests in the natural hollow



above the collarbone. Place padding under knot, if available. Tie the sling tightly enough to support the weight of the injured arm.

Types of Wounds

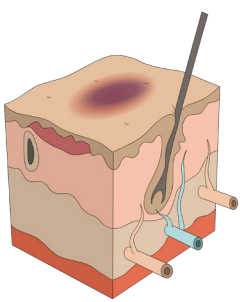
A wound is any damage to the soft tissues of the body. It usually results in the escape of blood from the blood vessels into surrounding tissues, body cavities or out of the body.

A wound can be either open or closed:

- **Open wound**—a break in the outer layer of the skin
- **Closed wound**—no break in the outer layer of skin but there is internal bleeding

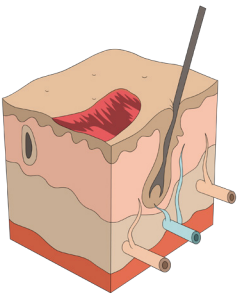
The aim in the care of wounds is to stop the bleeding and prevent infection. Although some bleeding may help to wash contamination from the wound, excessive blood flow must be stopped quickly to minimize shock.

Contusions or bruises



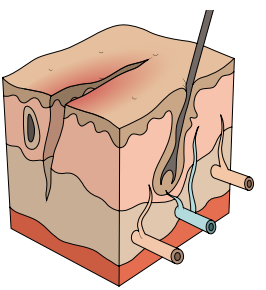
Contusions or bruises are closed wounds. The tissues under the skin are damaged and bleed into surrounding tissues, causing discolouration. A bruise may be a sign of a deeper, more serious injury or illness.

Abrasions or scrapes



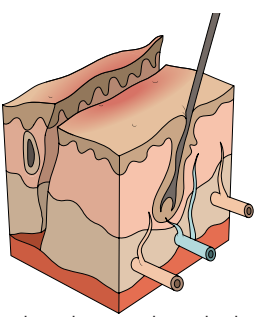
Abrasions or scrapes are open wounds where the outer protective layer of skin and the tiny underlying blood vessels are damaged. The deeper layer of the skin is still intact.

Incisions



Incisions are clean cuts caused by something sharp such as a knife.

Lacerations



Lacerations are tears in the skin and underlying tissue with jagged and irregular edges.